



Final Project Report

ICT 1209 - WEB TECHNOLOGIES

Department of Information and Communication Technology
Faculty of Technology
Rajarata University of Sri Lanka

Project Title: Diary Of Taste – Digital Recipe Book
Theme: Interactive Culinary Arts & Community Recipe Sharing Platform

Group Members:

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1. Project Overview

1.1.Purpose

The Diary of Taste is an interactive, responsive web-based digital recipe management platform. It allows home cooks, food enthusiasts, and culinary hobbyists to browse tested dishes, explore new cuisines, submit their own homemade recipes, and manage their cooking collection in an intuitive online portal. The system bridges community contributions with administrative quality control to maintain high content standards.

1.2.Target Audience

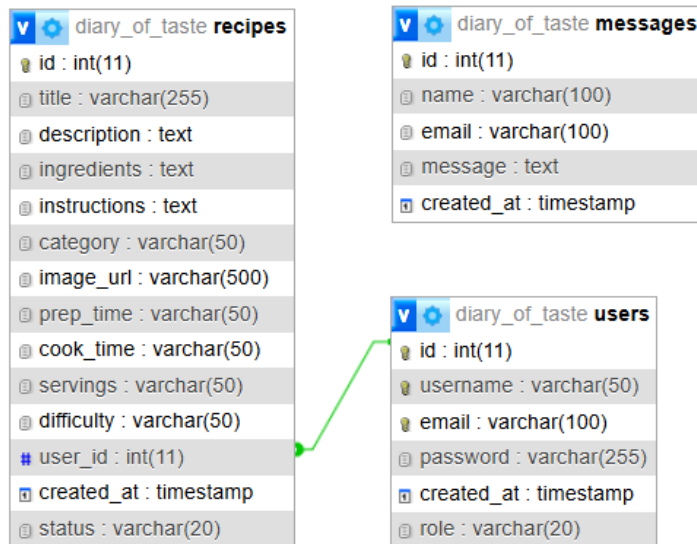
1. Home Cooks & Everyday Users: Individuals looking for curated recipes with step-by-step cooking instructions, ingredient checklists, preparation times, and difficulty indicators.
2. Community Contributors: Registered users who want to submit, manage, and edit their own recipes.
3. Platform Administrators: Site moderators responsible for reviewing pending community submissions, managing recipe visibility (Show/Hide), and responding to user feedback.

1.3.Key Features

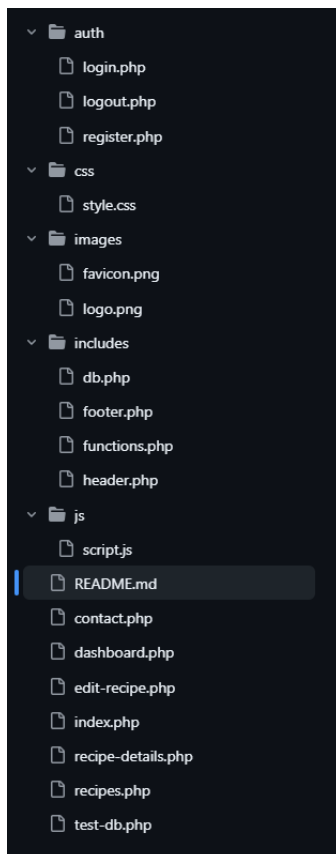
- User Authentication: Secure registration and login using PHP session management, BCrypt password hashing, and session regeneration to prevent session fixation.
- Recipe Exploration: Categorized browsing (Breakfast, Lunch, Dinner, Desserts, Drinks) with real-time category filtering.
- Hardened Recipe Search: Instant search by recipe title, description, or ingredients with SQL injection protection, wildcard escaping, and strict status isolation.
- Interactive Recipe Detail View: Full preparation breakdown, interactive ingredient checklists, difficulty and serving badges, related recipes, and a print-friendly view.
- Full Recipe CRUD with Ownership Protection: Authors can create, view, edit (edit-recipe.php), and delete their own recipes.
- Admin Review & Approval Workflow: User submissions enter a pending state; the admin can preview, approve (publish live), or reject them.
- Admin Visibility Control (Hide / Show): The administrator can toggle any recipe between Show (Live) and Hide (Unpublished) at any time.
- Contact Inquiries Management: Contact form submissions are saved to MySQL and are viewable/manageable by the administrator on the dashboard.

2. System Design

2.1.Database Schema (diary_of_taste)



2.2.Folder Structure



3. Implementation

3.1. User Authentication Module (auth/)

- Password Security: Passwords are never stored in plain text; they are hashed via PHP's `password_hash($password, PASSWORD_BCRYPT)` and checked using `password_verify()`.
- Session Protection: Upon successful authentication, `session_regenerate_id(true)` is immediately invoked to prevent session fixation attacks. User ID, username, and role are assigned to `$_SESSION`.
- Logout Flow: `logout.php` unsets session data, destroys the session using `session_destroy()`, and explicitly clears the session cookie (PHPSESSID).

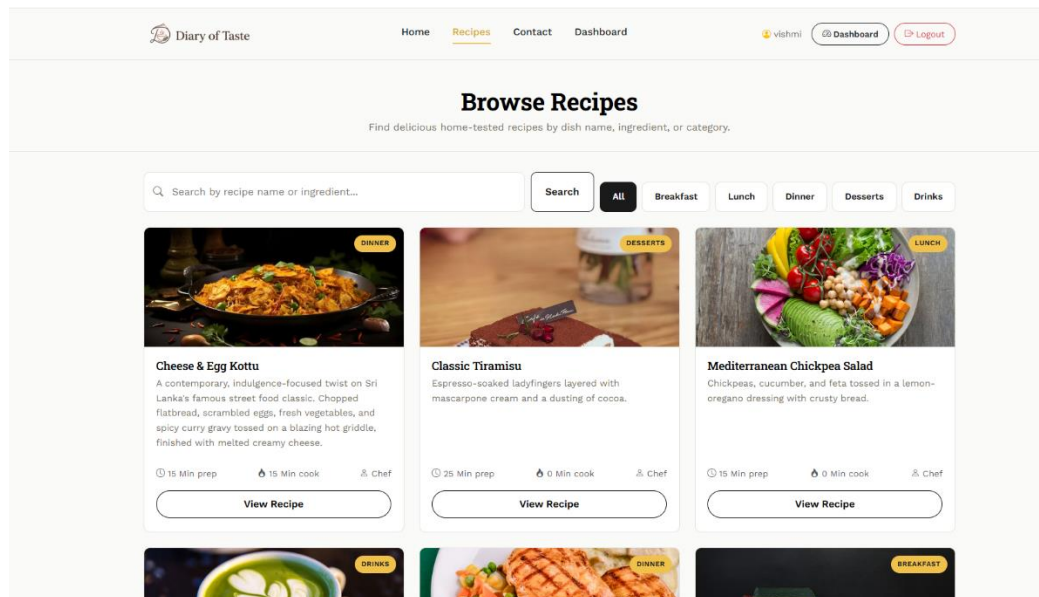
The image displays two web forms for user authentication. The left form is titled 'Create an Account' and includes fields for Username (example: e.g. chef_danuka), Email Address (example: e.g. you@example.com), Password (with a note 'At least 6 characters'), and Confirm Password (with a note 'Re-enter your password'). It features a 'Register Account' button and a link to 'Sign In here' for existing users. The right form is titled 'Sign In' and includes fields for Username or Email and Password, with a 'Sign In' button and a link to 'Register now' for new users. Below these forms is a navigation bar for 'Diary of Taste' with links to Home, Recipes, Contact, and Dashboard (highlighted), along with user links for vishmi, Dashboard, and Logout.

3.2. Recipe Catalog & Hardened Search (index.php, recipes.php)

- Dynamic Content: The homepage dynamically queries the 3 most recently approved recipes using `get_featured_recipes()`.
- Filterable Grid: Category pills (All, Breakfast, Lunch, Dinner, Desserts, Drinks) allow instant filtering.
- Search Hardening:

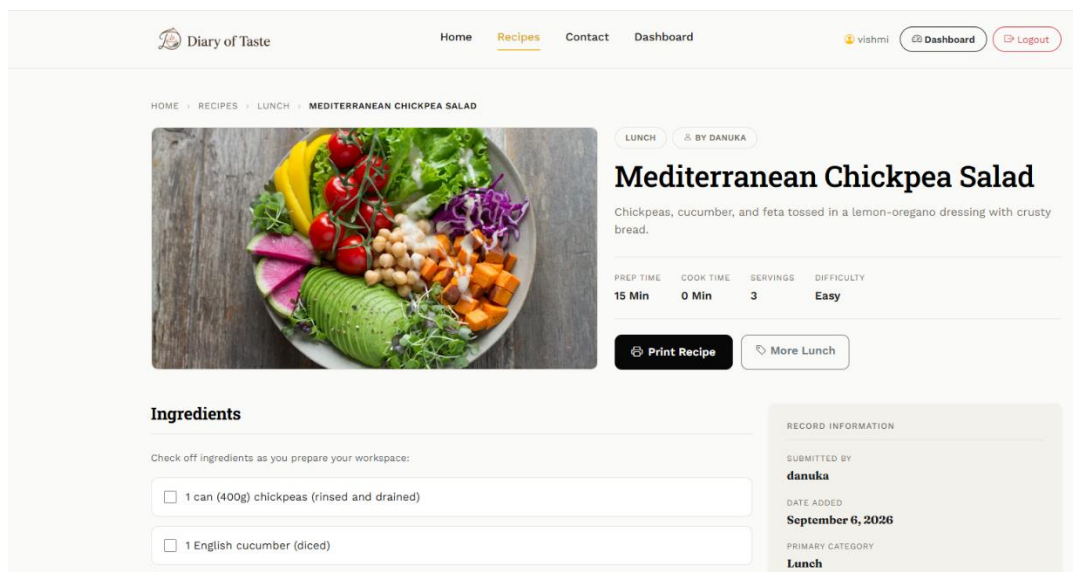
```
// includes/functions.php
$clean_search = mb_substr(trim($search), 0, 100);
$escaped_search = addslashes($clean_search, '%_');
$sql .= " AND (title LIKE ? OR description LIKE ? OR ingredients LIKE ?)";
```

This guarantees that SQL wildcards (`%`, `_`) cannot be abused, input is capped, and queries are strictly bound with PDO placeholders. Non-approved recipes are never returned.



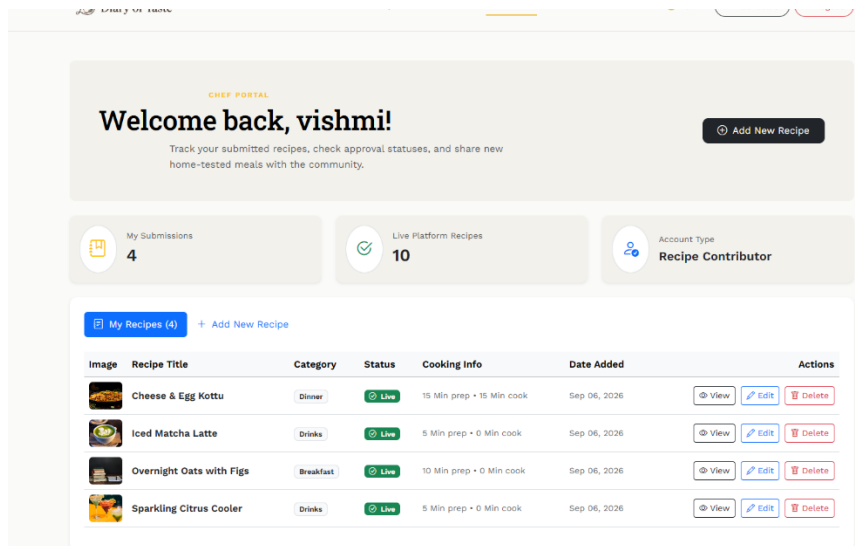
3.3. Recipe Detail View & Interactive Checklist (recipe-details.php)

- **Metadata Breakdown:** Shows category, author, prep time, cook time, servings, and difficulty level.
- **Interactive Ingredients Checklist:** Users can click checklist boxes next to ingredients as they cook.
- **Print Feature:** Triggers `window.print()` using CSS print styles to generate a clean recipe card without website navbars.
- **Preview Protection:** Non-approved recipes show an "Owner / Admin Preview" banner if accessed by the author/admin, or a "Recipe Currently Unavailable" notice to general visitors.



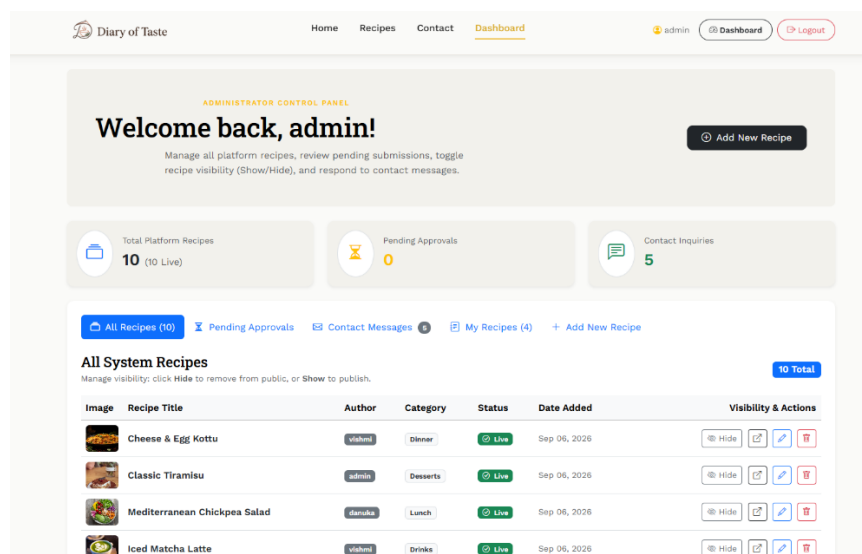
3.4. User Dashboard & Recipe CRUD (dashboard.php, edit-recipe.php)

- Submission Workflow: Users submit recipes via the Add New Recipe tab. Regular users see their recipe in Pending Approval state; admins are auto-approved.
- Author Recipe Editing (edit-recipe.php): Authors can modify their own submitted recipes. The system strictly checks `(int)$recipe['user_id'] === (int)$_SESSION['user_id'];` unauthorized attempts redirect with an error message.
- Safe Deletion: Authors can delete their own recipes, with a JavaScript confirmation dialog.



3.5. Admin Control Panel (dashboard.php)

- All Recipes Management (Show / Hide): Admin has a master list of all recipes and can toggle any recipe between Show (status = 'approved') and Hide (status = 'hidden') with one click.
- Pending Approvals Queue: Displays all newly submitted recipes with Preview, Approve, and Reject buttons.
- Contact Messages Panel: Lists all inquiries from contact.php with sender name, email (mailto:), message content, submission timestamp, and a Delete button.



4. Challenges and Solutions

	Challenge Encountered	Root Cause	Technical Solution Applied
1	Preventing Privilege Escalation (Role Tampering)	Malicious users could inject <code>role=admin</code> via POST payloads during registration or form submissions.	Hardcoded the default 'user' role, securely loaded session roles from the database, and protected admin actions with <code>is_admin()</code> .
2	SQL LIKE Wildcard & Information Leakage in Search	Unsanitized wildcards could trigger full-table matches or expose hidden recipes.	Enforced <code>status = 'approved'</code> , limited search to 100 characters, escaped wildcards, and sanitized HTML outputs.
3	Author-Only Access Control for Recipe Editing	Prevented unauthorized recipe modifications via URL ID manipulation.	Verified ownership during page load and update queries, automatically redirecting non-authors to the dashboard.

5. Individual Contribution Breakdown

Member Name	Index No	Areas of Contribution
H. A. D. B. Peiris	2764	- Designed the User Interface (UI) layouts and structural wireframes for the application. - Drafted the initial database schema (users, recipes, messages) in <code>database.sql</code>. - Sourced, collected, and optimized image assets and icons used throughout the user interface. - Populated the database with sample data (dummy users and test recipes) for initial system testing. - Assisted in compiling, writing, and formatting the final project report and documentation. - Developed the core User Authentication interfaces and routing, including the registration, login, and logout modules (<code>register.php</code>, <code>login.php</code>, <code>logout.php</code>).

R. D. W. M. V. A. Wijekoon	2804	• Authentication & Security: Built the complete user authentication system (with BCrypt hashing) and implemented strict author-only access controls for recipe editing. • Admin Dashboards & Workflows: Engineered the Chef and Admin dashboards, featuring recipe approval workflows, instant visibility toggles, and a contact message inspection panel. • Recipe Catalog: Developed the frontend recipe exploration page with category filtering and SQL injection-safe search logic. • Contact System: Implemented the full Contact Us module with server-side validation and database persistence. • Frontend & UI: Authored all responsive CSS styling and built the final visual UI components across the application.
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6. Setup Instructions (Running via XAMPP)

Step 1: Clone or Copy the Repository

Place the project directory into your XAMPP web root folder:
C:\xampp\htdocs\ict1209-digital-recipe-book\

Step 2: Start Apache and MySQL

Launch the XAMPP Control Panel.
Click Start for Apache.
Click Start for MySQL.

Step 3: Import the Database

Open your web browser and navigate to <http://localhost/phpmyadmin>.
Click the Import tab on the top navigation bar.
Click Choose File and select database.sql located inside the project directory (C:\xampp\htdocs\ict1209-digital-recipe-book\database.sql).
Click Go (or Import) at the bottom of the page. The script will automatically create the diary_of_taste database, establish all 3 tables (users, messages, recipes), and seed them with initial data.

Step 4: Verify Database Connection

Open your browser and visit: <http://localhost/ict1209-digital-recipe-book/test-db.php>
You should see: "Database Connected Successfully!" with table record counts.

Step 5: Launch the Application - Navigate to the homepage:

<http://localhost/ict1209-digital-recipe-book/index.php>

Step 6: Demo User Credentials

Username	Email	Password	Role	Permissions
admin	admin@diaryoftaste.com	password123	Administrator	Full platform control, approve/reject submissions, toggle Hide/Show on any recipe, read & delete contact messages
vishmi	vishmi@diaryoftaste.com	password123	User	Submit recipes, view personal submissions status, edit/delete owned recipes
luffy	luffy@example.com	password123	User	Submit recipes (enters pending state), edit/delete owned recipes

7. References

PHP Documentation Group (2024). PHP Data Objects (PDO) Manual. Available at:

<https://www.php.net/manual/en/book.pdo.php>

PHP Documentation Group (2024). Password Hashing Functions (password_hash, password_verify).

Available at: <https://www.php.net/manual/en/ref.password.php>

Oracle Corporation (2024). MySQL 8.0 Reference Manual: Foreign Key Constraints and Prepared Statements. Available at: <https://dev.mysql.com/doc/>

OWASP Foundation (2024). OWASP Top Ten Web Application Security Risks – Broken Access Control & Injection Prevention. Available at: <https://owasp.org/www-project-top-ten/>

Bootstrap Team (2024). Bootstrap 5.3 Framework Documentation. Available at:

<https://getbootstrap.com/docs/5.3/>

Mozilla Developer Network (MDN) (2024). HTTP Sessions, Cookies, and Web Security Best Practices. Available at: <https://developer.mozilla.org/en-US/docs/Web/HTTP/Cookies>